Electroencephalography and clinical Neurophysiology, 1989, 72: 184–187
Elsevier Scientific Publishers Ireland, Ltd.

EEG 02274

Short communication

Spherical splines for scalp potential and current density mapping

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(Accepted for publication: 12 September 1988)

Summary Description of mapping methods using spherical splines, both to interpolate scalp potentials (SPs), and to approximate scalp current densities (SCDs). Compared to a previously published method using thin plate splines, the advantages are a very simple derivation of the SCD approximation, faster computing times, and greater accuracy in areas with few electrodes.

Key words: Spherical splines; EEG or evoked potential mapping; Scalp current density

In preceding papers (Perrin et al. 1987a,b) we have shown how the 2-dimensional thin plate splines of Duchon (1976) can be applied to map EEG or scalp evoked potential data (SP) and to approximate scalp current densities (SCDs). In this note we describe the method we currently use which is based on the work of Wahba (1981, 1982) on spherical splines.

It is usually considered that a thin plate spline corresponds to the surface taken up by an infinite, thin plate constrained to pass through a finite number of known points. Instead of considering the deformation of a thin plate, Wahba has considered the deformation of a thin sphere constrained to pass through a finite number of known points.

The main advantage of these new interpolating surfaces is that they lead to simpler mathematical derivation and formulation of the SCD approximation. Consequently the computation time of a SCD map becomes identical to the computation time of the corresponding SP map.

To construct spatial maps of either scalp potentials or scalp current densities one must define precisely on which surface the scalp potential or scalp current densities are computed and what are the projections, first, between the scalp head surface and the surface of computation and, second, between the surface of computation and the plane on which the final map is drawn.

Usually the surface of computation of scalp potential maps is the projection plane. For SCDs we have used (Perrin et al. 1987b) a sphere as surface of computation. At least for SCDs it would be better if the surface of computation could be the scalp head surface.

In the following, the surface of computation for both SP and SCD will be the sphere and therefore two projections will have to be defined: one from the real scalp surface to the sphere, and one from the sphere to the plane.

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Projection from the real scalp surface onto the sphere

In the international 10–20 system one uses fractional distances between previously placed electrodes to locate new electrodes. For example the C_3 position is defined at half way (in curvilinear distance along the scalp surface) between C_z and T_3.

To project the scalp electrode positions on the sphere one follows on the sphere exactly the same procedure as that which was used on the scalp (Estrin and Uzgalis, 1969) with the following conventions: the spherical projection C_z of C_z is at the top of the sphere; the spherical projections F_{pz}, O_z of F_{pz} and O_z are diametrically opposite in the equatorial line, as are the projections T_3, T_4. If a certain percentage of the scalp curvilinear distance between two previously placed electrodes is used to place a new electrode in between, then the same percentage of the spherical arc length between the projections of the previously placed electrodes will be used to determine the projection, on the sphere, of the new electrode. By so doing, each electrode projection may be precisely located by its spherical coordinates, for instance: C_z (0, 0), C_3 (45, 180), O_z (90, 270).

Spherical interpolation

The second step consists of interpolating the potential data. Wahba presented two related spherical interpolation methods, one using spherical spline surfaces, the other pseudo-spherical spline surfaces. The difference between the two is that the mathematical expression of the latter depends on known func-

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tions (logarithms, square roots, etc.), whereas the former depends on an infinite series whose sum is not readily expressible in terms of simple known mathematical functions. We tested both and obtained slightly better results with the spherical spline interpolation method, which is presented here.

Let z_i be the potential value measured at the i^{th} electrode whose spherical projection will be denoted E_i. The value at a point E of the sphere, of the spherical spline U which interpolates the z_is at the E_is is given by:

$$U(E) = c_0 + \sum_{i=1}^{n} c_i g(\cos(E, E_i)), \tag{1}$$

where the c_is are solutions of:

$$\begin{cases} GC + Tc_0 = Z, \\ T'C = 0, \end{cases} \tag{2}$$

with

$$T' = (1, 1, \ldots, 1), \quad C' = (c_1, c_2, \ldots, c_n),$$

$$Z' = (z_1, z_2, \ldots, z_n), \quad G = (g_{ij}) = \left(g\left(\cos(E_i, E_j)\right)\right).$$

$\mathrm{Cos}(E_i, E_j)$ denotes the cosine of the angle between the electrode projections E_i and E_j.

The function g(x) is defined as the sum of the following series:

$$g(x) = \frac{1}{4\pi} \sum_{n=1}^{\infty} \frac{2n+1}{n^m (n+1)^m} P_n(x), \tag{3}$$

where P_n is the n^{th} degree Legendre polynomial. Depending on the constant m chosen (it must be an integer greater than 1) one obtains a family of functions U. With a methodology identical to the one previously used (Perrin et al. 1987a), all values of m between 2 and 6 were tested on simulations and the value of m equals 4 was chosen (Fig. 1). The function g(x) was tabulated once and for all, for 2000 values of x regularly disposed between -1 and 1. If (x_E, y_E, z_E) and (x_F, y_F, z_F) are the cartesian coordinates of E and F, and assuming a unit sphere (radius one) then

$$\cos(E, F) = 1 - \frac{(x_E - x_F)^2 + (y_E - y_F)^2 + (z_E - z_F)^2}{2}. \tag{4}$$

It should be noted that $P_n(x)$ can be computed via the recurrence relation:

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x),$$

with $P_0 = 1$ and $P_1 = x$. It should be further noted that, for m equals 4, the sum of the first 7 terms of the series is sufficient to obtain a precision of 10^{-6} on g(x) and that the resolution of (2) can be done, for example, by subprograms SSPFA and SSPSL of the LINPACK library (Dongarra et al. 1979).

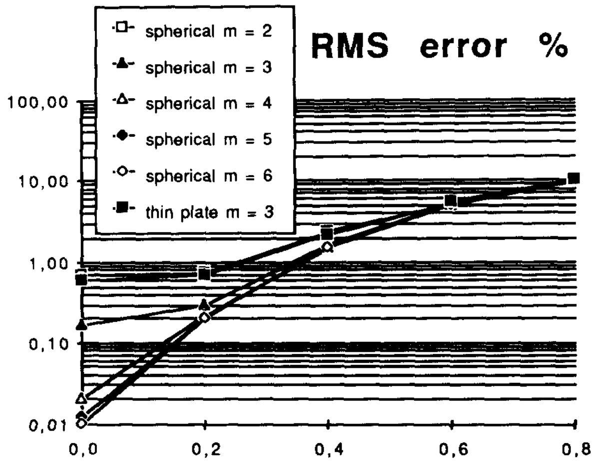

Fig. 1. RMS errors (evaluated at 5783 points regularly disposed on a hemisphere) between 'true' potential values obtained from a 3-concentric shell head model and interpolated potential values obtained from the 'true' values at the 19 electrode locations of the international 10–20 system. But for those of eccentricity 0, each point corresponds to the mean value of the RMS errors of 15 dipoles of same eccentricity but differing orientations and positions.

Projection from the sphere onto the plane

Having computed the c_is, the third step consists of evaluating the interpolated potential at each map pixel. This necessitates the choice of a projection from the sphere onto the plane. As before (Perrin et al. 1987b), a radial projection has been used. Thus, each map pixel center $M'(x', y')$ has its back projection $M(x_M, y_M, z_M)$, on the sphere, determined by:

$$\begin{cases} x_M = kx', \\ y_M = ky', \\ z_M = \sqrt{1 - x_M^2 - y_M^2}, \end{cases}$$

with

$$k = \frac{\sin(\theta)}{\theta} \quad \text{and} \quad \theta = \sqrt{x'^2 + y'^2}.$$

Knowing M, the quantities $\cos(M, E_i) = e_i$ are easily calculated by (4) and then $g(e_i)$ is linearly interpolated between the two nearest values of the g table. Having the $g(e_i)$s and the c_is, the value of the potential is given by (1).

Scalp current density estimations

Due to the classical property that the 2-dimensional spherical laplacian of the Legendre polynomials is a multiple of the same Legendre polynomial:

$$\Delta P_n = -(2n+1)P_n,$$

the expression of the current density C(E), which is proportional to minus the 2-dimensional spherical laplacian of the potential (Perrin et al. 1987b), becomes quite simple:

$$C(E) = \sum_{i=1}^{n} c_i h(\cos(E, E_i)), \tag{5}$$

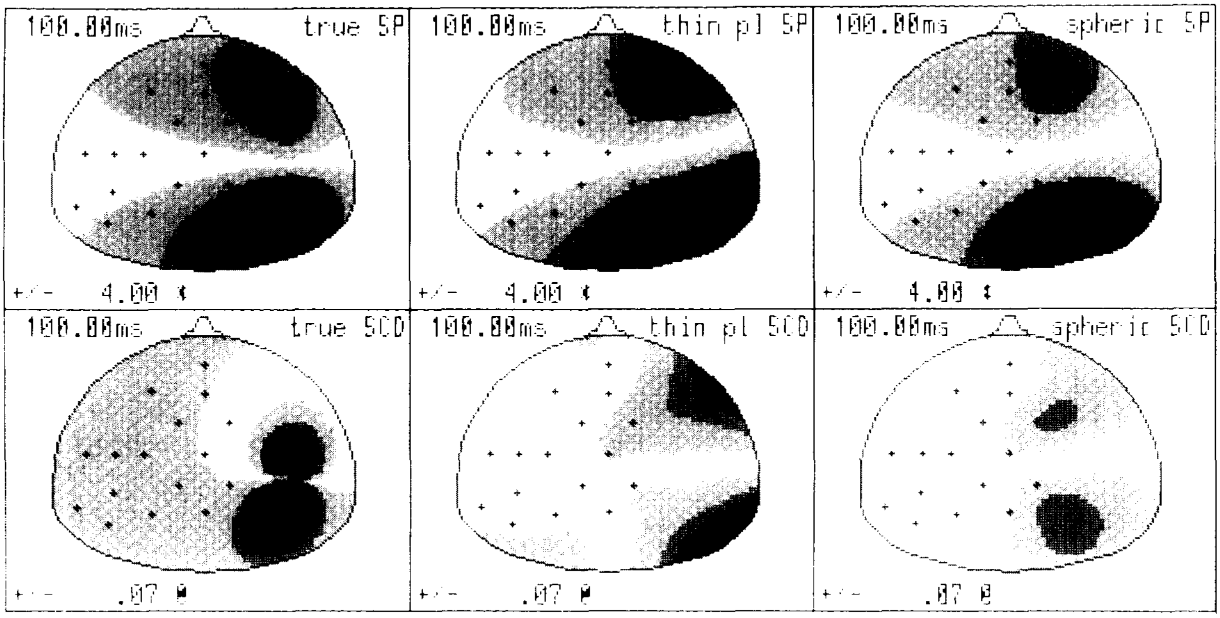

Fig. 2. Above: scalp potential maps. Below: corresponding scalp current density maps, obtained from a 3-concentric shell head model with a tangential current dipole located in the right part of the sphere, whereas the 16 electrodes are on the left part. From left to right: the 'true' maps, computed from the model at each pixel; the thin plate interpolated maps; the spherical interpolated maps.

with

$$h(x) = -\frac{1}{4\pi}\sum_{n=1}^{\infty}\frac{(2n+1)^2}{n^m(n+1)^m}P_n(x).$$

Thus if one tabulates h(x) in the same way one has tabulated g(x), all one has to do to obtain C(E) is to replace, in the third step, g(x) by h(x) and eqn. (1) by eqn. (5). Thus the computation time of C(M) will be exactly the same as that of U(M). To appreciate the improvement in simplicity one should compare the above eqn. (5) to the corresponding eqn. (5) of the previously published method (Perrin et al. 1987b). This corresponds to a time improvement of 41%. To tabulate h(x), when m equals 4, with a precision of 10^{-6}, 20 terms are sufficient.

Comments

In simulations, thin plate splines and spherical splines give results which are quite similar in areas well covered by electrodes, while they differ somewhat in undersampled areas, where spherical splines seem to have a slight advantage over thin plate splines (Fig. 2).

As Harder and Desmarais (1972) have already noted, the same computational scheme can be used to obtain approximations instead of interpolations, by choosing a positive lambda and solving:

$$\begin{cases}(G+\lambda 1)C+Tc_0=Z,\\ T'C=0.\end{cases}$$

Statistical methods are available to choose the optimal value of lambda (Wahba and Wendelberger 1980). This may be useful when potential measurements are noisy. Indeed, an interpolation surface is constrained to pass through the potential values at the electrode sites, whereas an approximated surface may smooth some irregularities due to the noise by taking it into account in the computation procedure. Moreover, the same procedure gives confidence intervals for the approximated map (Wahba 1983).

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